

Please pick only one of the two cases below to work on, the engineering case or the research case, not both. We are more interested in your way of thinking than in you providing all the right answers. The questions are explicitly open ended, and there is no right answer. If anything is unclear, please reach out to Levin.

ENGINEERING CASE

A small environment with a broken reward

Build a small, locally runnable environment for one concrete task from manufacturing or industrial engineering (machine scheduling, order picking, quality inspection, replenishment; your choice). Success must be checkable automatically, without a human judge.

- `reset()`, `step(action)`, observation, reward, and a separate verifier that reads only the final state
- at least three task variants generated by your code
- a simple baseline agent (heuristic or search, no training needed) with its success rate over all variants
- verifier tests: initial state scores 0, correct solution scores 1, invalid action or directly written goal state scores 0
- one documented exploit: a naive reward that a wrong solution can satisfy, your fix, and the test that proves the fix

What we look for: scope, choice of task, and what a buyer of this environment would need next.

RESEARCH CASE

When does real operational data beat generated data?

Labs can generate enterprise data at will: procedurally built companies, LLM-written records. Pick one narrow (!) part of this question, state a hypothesis, and test it yourself.

- one hypothesis about a specific way real operational data differs from generated data and why that matters for training or evaluating agents (for example: error structure, rare cases, cross-document consistency, drift over time); make it falsifiable
- a small experiment that tests it: public real data, or a small sample you collect, against generated data of the same shape
- a result, honestly reported, including the case where the hypothesis does not hold
- what your result does not show, and what it would take to test the general question
- a short research paper of at most four pages (question, hypothesis, method, result, limits)

What we look for: whether the hypothesis is sharp, whether the test could have failed, and whether you can tell what real data adds from what it merely seems to add.

DEADLINE

One week after you receive this sheet. Send us a link to a repository, public or shared with us.

ZERO COST

This must not cost you anything. No GPU hours, no API credits, no paid licenses (besides what you already have anyways).

TOOLS

Any language, any tools, any public source. Any sort of AI assistant for coding or whatever you want are explicitly allowed. Full autonomy on scope and approach.

DELIVERABLE

A clean repository that runs with one command or a short research paper, respectively, and a note of at most one page (decisions, results, limits, next step).